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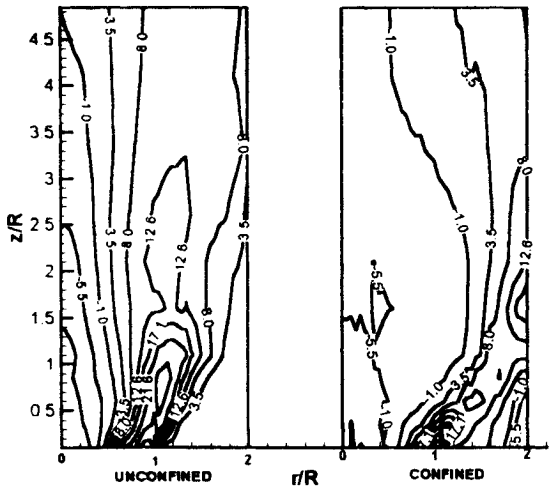


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*Modern industrial gas-turbine spray combustors feature multiple swirlers and distributed fuel injection for rapid mixing and stabilization. The present paper is one of a sequence of papers that aim to study the flow field of such combustor, the related combustion characteristics and their control. The present paper focuses on experimental investigation of the velocity flow field downstream of a Triple Annular Research Swirlers (TARS). Multiple combinations of swirlers were tested in cold flow under atmospheric conditions with and without confined combustion chamber. The experiments showed that a central recirculation zone, an annular jet with internal and external shear layers characterize the flow field downstream of TARS. The Central Toroidal Recirculation Zone (CTRZ) region is axisymmetric but the jet contains imprints of the internal flow and has some nonaxisymmetric features. Comparison of experimental results with CFD simulations are discussed. Numerical results confirmed these observations and showed that the asymmetry relates to the effect of the internal swirling vanes on the jet flow. Compared to unconfined case, flow with confined tube shows an enlarged CTRZ region and a recirculation region in the expansion corner and reduced concentration of turbulence intensity in the jet region.*

## Introduction

Many modern combustion systems are characterized by multiple annular counter-rotating [1] and co-rotating [2] axial swirlers with distributed fuel injection points. This design results in efficient combustion with extremely low emissions and improved combustor performance. These systems are usually operated near the lean flammability limit where they may exhibit combustion instability [1]. The various mechanisms that may trigger combustion instabilities have been studied extensively in dump combustors, bluff body stabilized system and single swirler combustors [3,4]. Flow related instabilities, equivalence ratio fluctuations, acoustic coupling and unstable spray dynamics were identified as the dominant mechanisms that drive thermoacoustic oscillations when Rayleigh's instability criterion [5] is satisfied. Studies of the fluid dynamics of multi-swirlers systems, their mixing characteristics, the flow instabilities and the potential driving mechanisms of combustion instabilities are quite limited. To achieve optimum combustion performance while minimizing the propensity to develop thermoacoustic instabilities it is important to study the cold and hot flow patterns in a geometry that simulates the design of modern multi swirl combustor systems.

The Triple Annular Research Swirler (TARS), which is the topic of the present investigation, features three swirling air passages and distributed fuel injection pattern. The nozzle is developed by Delavan Gas Turbine Products (a division of Goodrich Aerospace) in cooperation with General Electric Aircraft Engines (GEAE). The purpose of the study is to investigate the effect of the internal design of the fuel injector on the mixing pattern in the combustion chamber.

Numerous efforts have been devoted to study swirling flows in various combustion systems such as gas turbine engines and diesel engines. The swirling flows are used to improve and control the mixing process between fuel and oxidant streams and in order to achieve flame stabilization and enhanced heat release rate [6]. Velocity field characteristics of swirling flow combustors have been investigated extensively but limited to specific geometry and were primarily focused on swirling jet flows [7] or on the interaction of two co- or counter swirling streams [8]. One of the main features that are usually observed in swirling flows is the formation of a Center Toroidal Recirculation Zone (CTRZ) around the axis of the jet. This

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CTRZ is used to stabilize the combustion process in a compact region within the combustion chamber. Experimental results showed that the CTRZ is a quasi-axisymmetric bubble developed by vortex breakdown which is associated with swirling flow exceeding a certain swirling strength [7,8]. Multiple factors, including inlet conditions and geometry [9], tangential velocity profile and axial velocity [10] and downstream conditions [11] affect the process of vortex evolution and breakdown process.

In a complex geometry such as TARS in which several co-swirling or counter-swirling flows interact, the vortical evolution and breakdown are complex and detailed velocity measurements are necessary to resolve the physical processes that involve mixing and interaction of three distinct air streams.

While detailed measurements are possible downstream of the swirler, the internal passages are more difficult to probe. Numerical simulation is preferred for investigating the internal flow structure. The numerical results can be compared at the nozzle exit with the experiments for validation. Accurate numerical simulation can also be used as an effective design tool of complex combustor geometry. Due to the three dimensionality and periodic character of the swirling flow, accurate prediction is challenging and requires adequate approach. Spall and Gatski [12] have used RANS to simulate velocity/vorticity field and predict the turbulent vortex breakdown. Their simulations were compared with experimental results.

The present paper compares the flow field of four different swirler combinations under the same operating conditions. Simulation results will be presented and compared with experimental data at the same locations in the flow field. The information obtained here will subsequently be used to study the fluid dynamics effect on the flame structure in a multiple swirl stabilized spray combustor and to develop passive/active control strategies for improved performance, such as acoustic instability abatement and emission reduction in industrial combustors.

## Experimental Setup

The research was conducted in a cold flow combustor test rig simulating the exact geometry of the hot combustion rig. Schematic drawing of this facility is shown in Figure 1 (units in inches).

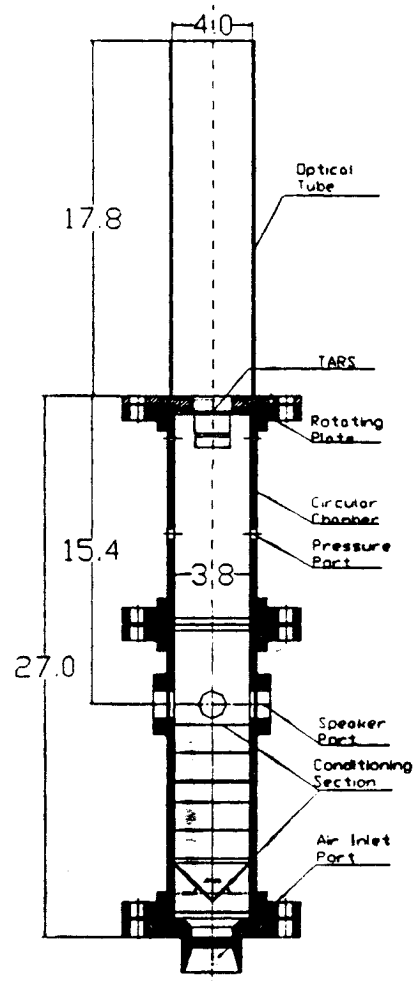


**Figure1: Schematic drawing of cold flow test rig for LDV measurement of different TARS combinations**

This test rig is set up vertically and the air is introduced at the bottom, conditioned through air conditioning section which is composed of perforated cone, screens and honeycomb, settled in 3.8" diameter circular chamber, fed into a Triple Annular Research Swirlers (TARS) and exits to the atmosphere. The pressure drop across TARS is 4% and at atmosphere pressure it is about 0.6 psi. TARS features three separate air flow passages and independent liquid fuel supply lines which can be controlled separately (Fig. 2).

In order to simulate thermoacoustic instability during combustion, four 75 watts acoustic speakers are installed circumferentially at upstream of the nozzle inlet. Air passes through the nozzle, premixes with the fuel spray, attains strong swirl, and then suddenly expands into 4" diameter cylinder chamber, which has full optical access for laser diagnostics, including

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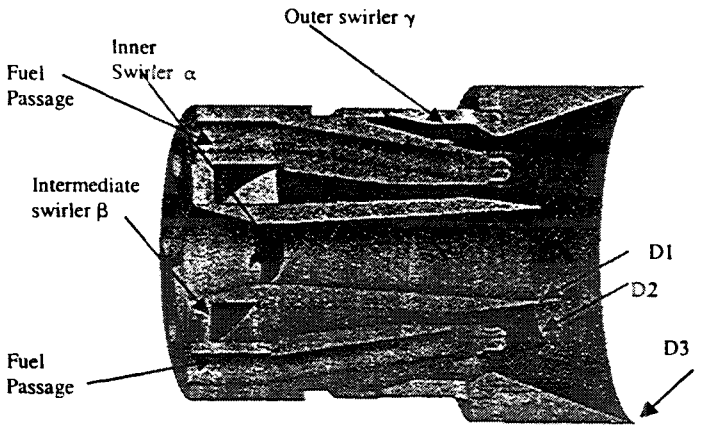


Figure 2: Cross Sectional Drawing of TARS. (D1=0.55";D2= 1.05";D3= 2")

PIV and LDV. Data presented in this paper is with no acoustic forcing.

The time averaged velocity field of air flow downstream of the TARS, including axial and tangential mean and turbulent velocity components were measured using a 2-D TSI Model 3200L/P Phase Doppler Particle Analyzer employing a forward scatter mode with a scatter angle of 30 degree. In order to measure the axial and tangential velocity components simultaneously, the test rig was mounted on a 2-D transverse system, Velmex Model 34D-9109A. The transverse system, controlled by Labview software, was used to move the test rig stably along the horizontal and vertical directions relative to the LDV beams. The step length and motor speed were controlled by the software.

An argon-ion laser system made by Ion Laser Technology, Inc. provided the coherent laser source with wavelengths of 514.5 and 488 nm. The system was configured as a 2-D LDV, using the green beam (514.5 nm) to measure the axial velocity parallel to the axis of the nozzle and the blue beam (488 nm) to measure the tangential velocity in the horizontal plane perpendicular to the nozzle axis. The focal length of the transmitter was 500 mm/250mm and the receiver was 300 mm. When the test rig traversed horizontally along a radius that was aligned with the axis of the transmitter probe, axial and tangential velocity components were measured along this radius. By rotating the TARS mounting plate, data along a set of radii was obtained and was used to construct the velocity distribution in a cross-sectional plane. By measuring along the same radius at different vertical positions, the velocity distribution in a streamwise plane was obtained.

To improve the signal quality and data rate of the LDV measurements, a seeding system was built to seed the entire airflow through the nozzle. The seeder included a ROSCO 1500 Fog Machine which was installed inside a high pressure vessel. Compressed air mixed the fog inside the vessel and delivered it into the test rig.

## Results and Discussion

Time averaged mean and turbulent velocity data without a confined combustion chamber was used to compare the basic characteristics of the flow field for the different multiple swirler combinations listed in Table 1 (Fig. 3). The

| | Swirler 304545 | Swirler 3045c45 | Swirler 304545c | Swirler 504545 |
|---|---|---|---|---|
| γ (°) | 30 | 30 | 30 | 50 |
| Number of Vanes | 8 | 8 | 8 | 8 |
| β (°) | 45 | 45c | 45 | 45 |
| Number of Vanes | 4 | 4 | 4 | 4 |
| α (°) | 45 | 45 | 45c | 45 |
| Number of Vanes | 4 | 4 | 4 | 4 |
| Swirl. No. | 0.39 | 0.22 | 0.36 | 0.40 |
| M (kg/s) | 0.040 | 0.046 | 0.041 | 0.046 |

Table 1: Combinations of TARS swirlers included in the study. "c" denotes counter-clockwise rotation relative to the incoming flow.

different swirlers in the table are characterized by the corresponding vane angles of the inner, intermediate, and outer air passages. The table shows the swirl number, which is calculated using equation (1) and the mass flow rate through the nozzle calculated using equation (2). The flow was ducted through the TARS nozzle and exited to ambient atmosphere. The outlet of the TARS was located at the inlet dump plane of the combustion chamber. This location was defined as z/R=0, where R is the radius of the TARS outlet.

$$S = \frac{\int_0^R \rho U W r^2 dr}{R\int_0^R \rho U^2 r dr} \qquad (1)$$

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$$m = \int_0^R \rho U 2\pi r dr \qquad (2)$$

Where, U, W are axial and tangential velocity components respectively, ρ is density and R is the radius of the circular jet.

**Velocity Mapping of Swirler 504545**

A complete mapping of the axial and tangential velocity components was performed in a plane perpendicular to the axis at z/R = 0.1 and in the streamwise plane which included the axis of the nozzle. The mean and RMS axial and tangential velocity components mapping at z/R=0.1 for swirler504545 combination is shown in Figure 4. The following flow structures are observed in this figure:

1) The axial velocity mapping shows a central axisymmetric recirculation zone that was formed around the axis with a diameter of about 1R at this plane.

2) Close observation revealed that at z/R = 0.1 the flow region outside of the CTRZ is not fully axisymmetric and several peaks can be discerned near r/R=0.9 in the mean and r/R=0.6 in RMS axial velocity mapping and in the tangential velocity mapping.

These peaks may be correlated to the number of vanes in the inner, intermediate and outer swirlers which were 4, 4, and 8, respectively. At the location of z/D=0.05 the non-axisymmetric flow induced by the swirler vanes was not fully mixed to produce uniform flow and some locations (corresponding to the vane geometry) have larger velocity components than others.

Similar behavior is shown in Figure 5, which depicts the velocity mapping on a downstream cross-sectional plane at z/R=0.4. Distinct peaks at mean and RMS velocity mapping can be clearly observed.

The contour plots of Figure 4 were averaged circumferentially to produce radial profiles of the axial and tangential velocity components as shown in Figure 6 for swirler 504545 at z/R=0.1. The locations of the peak velocity of the mean and tangential components are at r/R=0.9 and r/R= 0.95 The peaks of the RMS velocity of the two components are at nearly the same locations at r/R=0.6 where the shear layer of annular jet is located. The CTRZ has a radius of r/R=0.5 which is primarily contributed by the inner swirler and partially by the intermediate swirler.

The mean and RMS axial and tangential velocities in the streamwise plane for this swirler are shown in Figure 7. This plane extends from z/R=0.1 to z/R=4.7. The negative region of mean axial velocity near the centerline shows that the region of the center recirculation zone extends to about z/R = 5 (extrapolated from the data). The momentum of the strong airflow in the annular jet is gradually reduced as the jet merges into the main stream at about z/R=2. High magnitude mean tangential velocity is confined to a small region near the nozzle exit. A secondary region of increased level tangential component is centered at about z/R=1.5. This amplified tangential velocity may be related to a circumferential flow instability mode. This phenomenon will be further investigated by PIV and hotwire to capture the unsteady nature of the flow. The RMS mapping shows that the multiple swirling flows mix well before reaching the z/R=2 plane.

**Comparison of Different Swirler Combinations**

The circumferentially averaged mean and turbulent axial and tangential velocity components were measured at an axial distance of z/R =0.1 for the different swirler combinations listed in Table1 (Figure 8). The important flow features that can be discerned in these figures are:

1) The counter-rotating sense of intermediate swirler causes significant reduction in the diameter of the recirculation zone from r/R=0.55 to r/R=0.35 while reducing the peak axial velocity and spreading the peak over a wider range of radial distance. It seems that the intermediate counter-clockwise rotating swirling flow pushes the inner flow inside and mixes with the outer flow in the radial direction better than the other swirlers. The intermediate flow has significant effect on the inner flow field as can be seen from the RMS of both axial and tangential components that increase significantly in the inner flow region. This enhanced mixing suggests that counter rotating swirler will be a good choice to improve the inner region fuel atomization but may reduce the flame stabilization.

2) The swirling sense of the inner swirler has a negligible effect on the flow field because the inner flow is mostly fully developed. This can be seen from the nearly identical profiles of swirlers 304545 and 304545c.

3) The larger swirling angle of the outer swirler (504545) compared to swirler 304545, leads to higher magnitude of both the axial and tangential velocity components. There was no apparent effect of this increased swirl on the axial or tangential RMS levels.

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4) For case3045c45 and case4545c, the flow region between inner and intermediate swirler are subject to same counter-rotating shear layer but the resulted flow are significantly different. One possible reason is that the outer shear layer, which is formed by the air through outer and intermediate air passages, puts some effects into this inner shear layer region.

**Comparison of Data With /Without Confined Tube**

Data for swirler 504545 with and without a confining tube that simulates the combustion chamber were taken under the same operating condition for streamwise plane (Figure 9) and cross section plane (Figure10). Figure 9 (a) and (c) show that the jet and CTRZ are significantly enlarged by the tube both in radial and axial direction There is an external recirculation zone, which is located at the corner of the expansion tube. Another important observation is that the maximum axial velocity area, which is spread between r/R=1~1.2 and z/R=0~1.2 in the unconfined case, tends to be damped immediately after exiting the nozzle. The turbulent axial velocity component (Fig. 9b) is reduced and is more uniformly distributed in the center region of the jet. This suggests that the confined tube acts damping the turbulence by creating a larger uniform CTRZ. This cold flow observation may have significant effect on the combustion stability in reacting flow studies. Circumferential averaged velocity profiles in Figure 10 confirm this observation quantitatively by showing the magnitude reduction for confined case in the mean and RMS velocity compared with unconfined case.

**Comparison of experimental data with CFD simulation**

CFD simulations based on the same operating conditions were performed for swirlers 304545, 3045c45 and 304545c by using Fluent 5.5 3D solver and realizable $k$-$\varepsilon$ model. The comparison of the circumferentially averaged mean axial and tangential velocities obtained from CFD simulations and those obtained from experimental data, for swirler 304545 at z/R=0.25 is shown in Figure 11. The magnitude and peak location of tangential velocity agrees well with experimental measurements. The features of the axial velocity profile calculated using CFD follow closely those of the experimental data, except the location of the peak jet velocity which is shifted from r/R=1 (experimental) to r/R=0.75 (CFD). The CFD simulations of the entire flow field at the exit of the nozzle (not shown here) showed a region of separated flow at the boundary between the intermediate and outer air passages. The flow separation predicted by the simulations caused the annular jet to shift inwards resulting in inner displacement of the velocity peak. No such separation was observed in the experiments which showed that the flow remained attached to the wall between the inner and intermediate swirlers. This discrepancy between the simulations and experiments is the result of the inability of the CFD to predict the proper behavior of a turbulent swirling flow near a diverging wall.

The internal flow in the TARS swirler was simulated using CFD (Fig. 12). The series of figures at different axial cross-sections (z/R locations) show the flow evolution process. The initial contours at z/R=-0.4 show the imprint of the eight vane passages. As the flow moves towards the exit, the eight streams merge, and at the exit plane (z/R=0) there were four high velocity regions, similar to those observed in the measurements presented in Fig. 4. The four spots of the intermediate flow seem to split out from the axisymmetric flow in the inner passage.

## Conclusion

LDV measurements were performed on a Triple Annular Research Swirler (TARS) in cold flow test rig that simulates an identical combustion rig. Axial and tangential velocity mapping of the flow in different cross sectional planes and along a centerline streamwise plane revealed several important flow structures in the triple annular swirling flow with co-and counter swirling cases. An axisymmetric recirculation zone was formed in the center of the flow but the outer regions showed some non-axisymmetric features due to incomplete internal mixing. The strong RMS magnitude induced by the counter-clockwise intermediate swirler suggests that this configuration could be a good choice to obtain enhanced fuel and air mixing. However, the same configuration resulted in reduced central recirculation zone which may lead to flame destabilization.

The comparison between confined and unconfined results for swirler 504545 shows the significant effect of the confining tube on the swirling flow in terms of jet expansion and decay rate, enlargement of the CRTZ and reduced turbulent intensity which becomes more uniform in the center jet region.

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Steady state 3-D CFD simulations confirmed features that were observed in the experiment, including the formation of the central recirculation zone and the four lobed flow at the nozzle exit. The simulations provided information on the internal flow development from an initial region which is mostly affected by the vanes geometry to the external zone that is governed by the interaction between the three streams. Reasonable agreement between the velocity profiles predicted by the CFD and the experimental data suggest that the simulation approach used here can be an effective design tool to determine the flow field in fuel nozzles with complex geometry. Issues of wrong prediction of flow separation should be addressed. Additional measurement will be performed to resolve the dynamic nature of the flow field associated with the present complex geometry.

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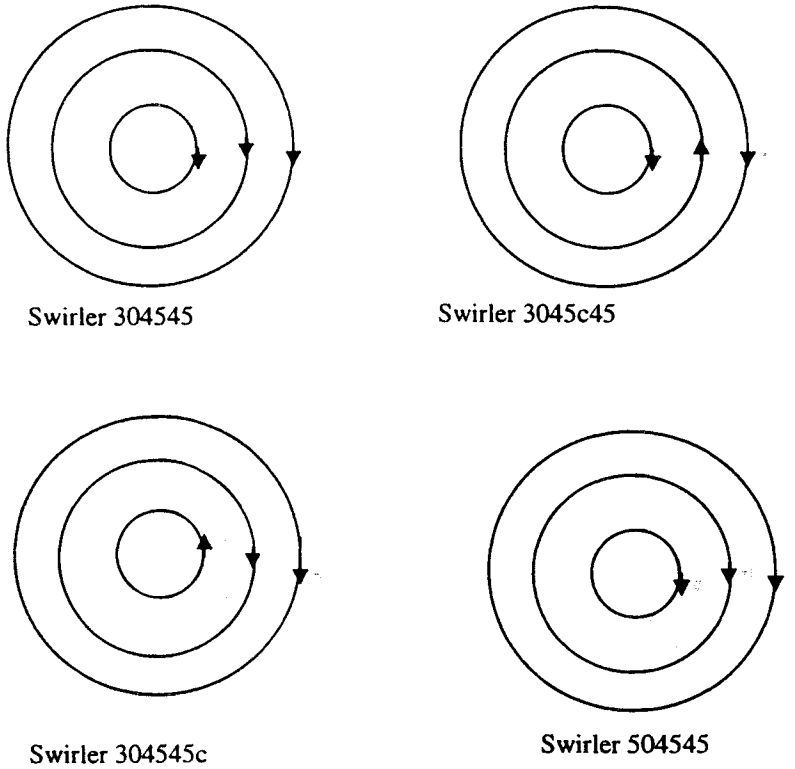


Figure 3: Schematic drawing of swirling flow direction for different swirlers

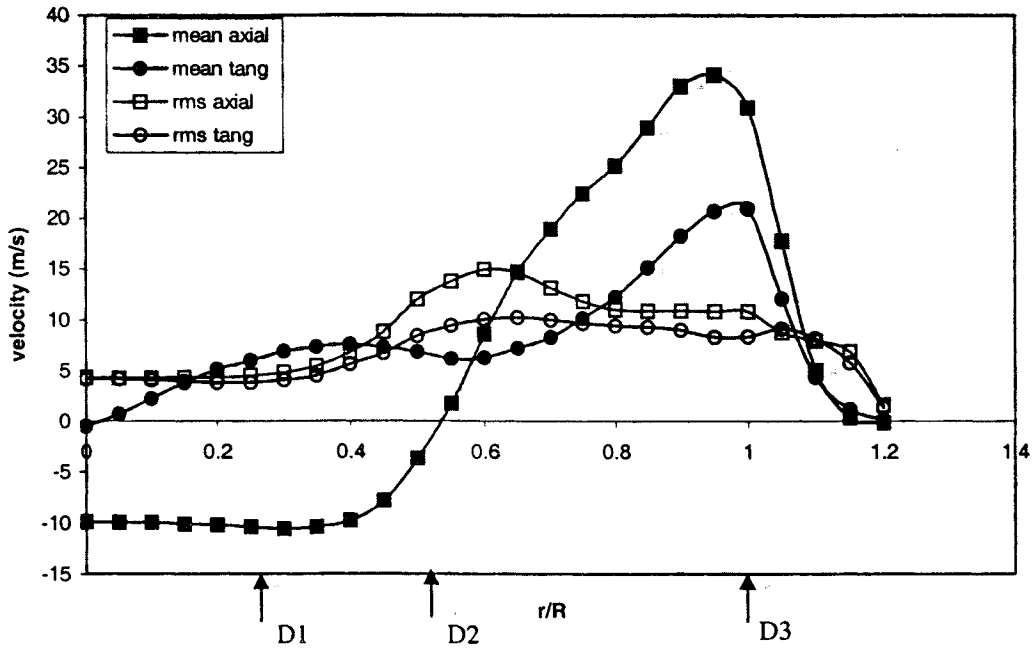


Figure 6: Circumferential Averaged Velocity Profile at z/R=0.1 for swirler 504545

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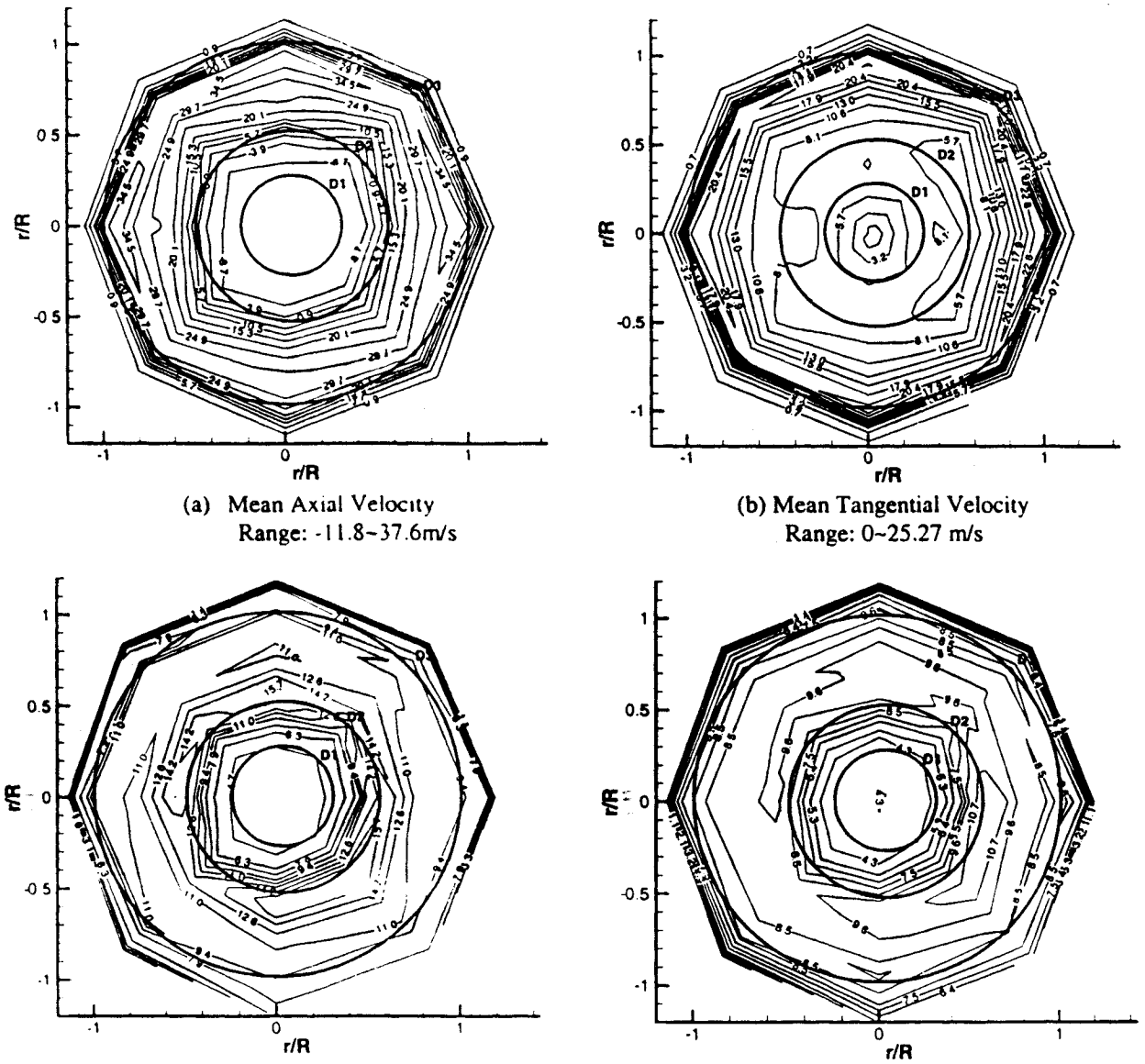


(a) Mean Axial Velocity
Range: -11.8~37.6m/s

(b) Mean Tangential Velocity
Range: 0~25.27 m/s

(c) RMS Axial Velocity
Range: 0~17.3 m/s

(d) RMS Tangential Velocity
Range: 0~11.7 ms/

Figure 4: Velocity Mapping of Swirler 504545 at z/R=0.1 cross sectional plane

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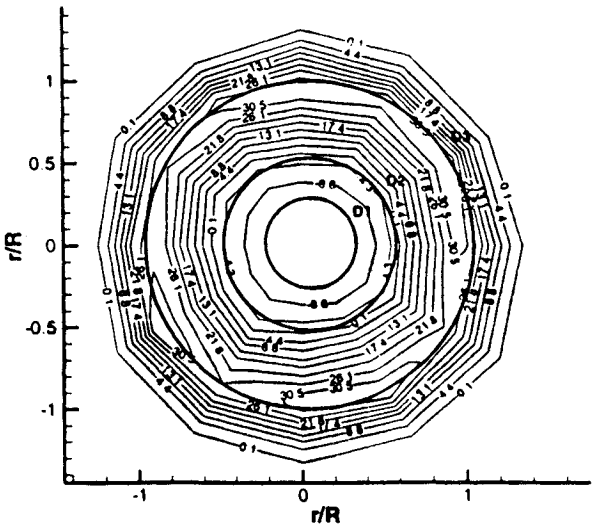
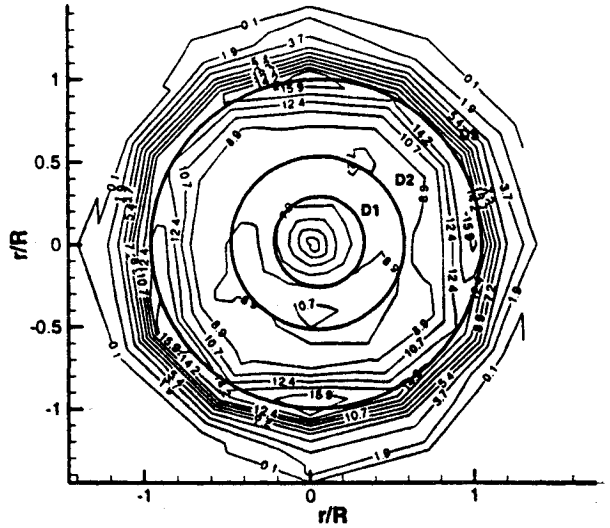
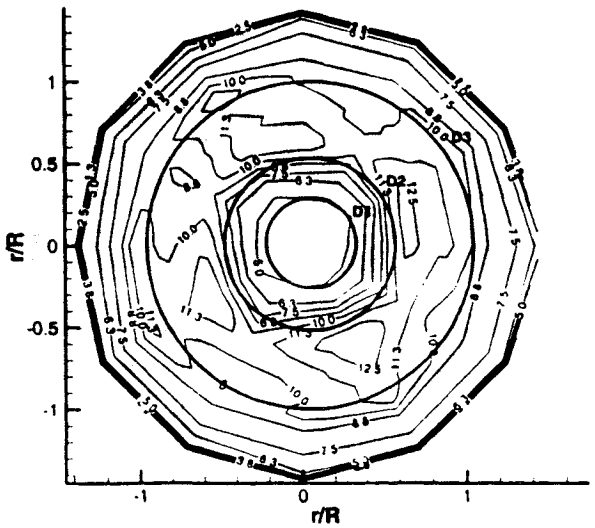


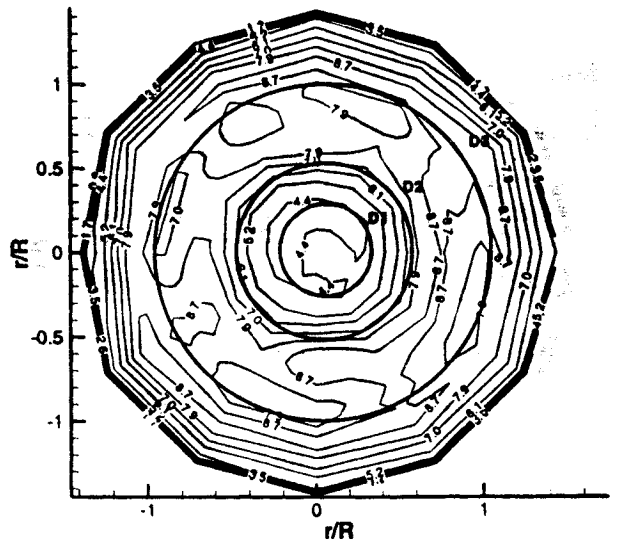

(a) Mean Axial Velocity
Range: -12.96~34.82m/s




(b) Mean Tangential Velocity
Range: 0~17 m/s




(c) RMS Axial Velocity
Range: 0~13.77 m/s




(d) RMS Tangential Velocity (m/s)
Range: 0~9.61m/s

Figure 5: Velocity mapping at z/R=0.4 for swirler 504545

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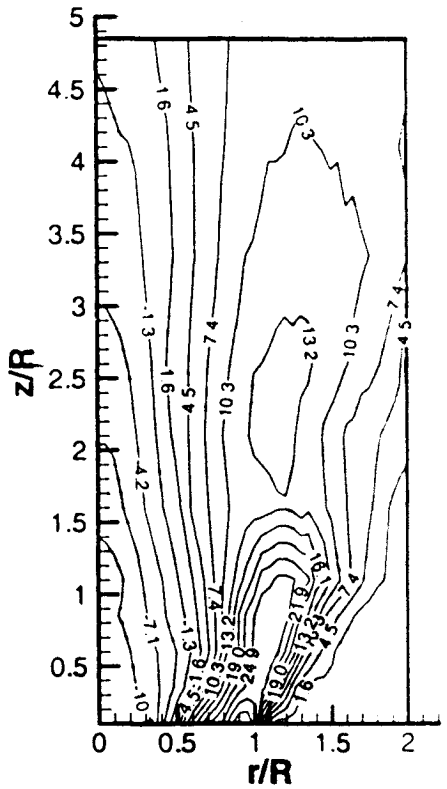
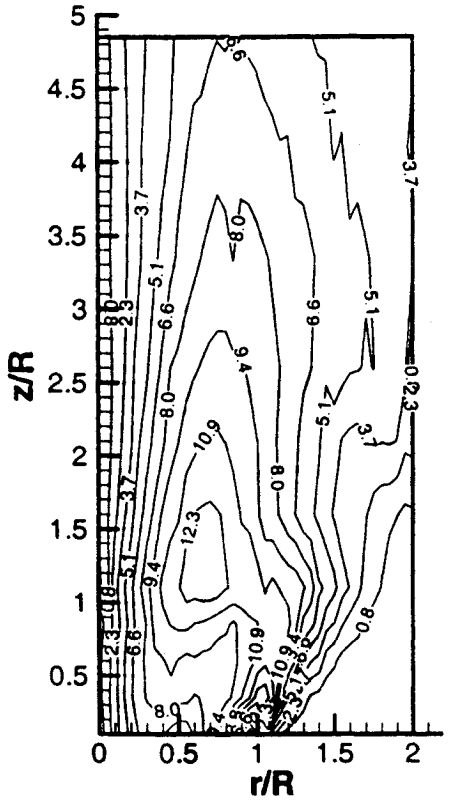
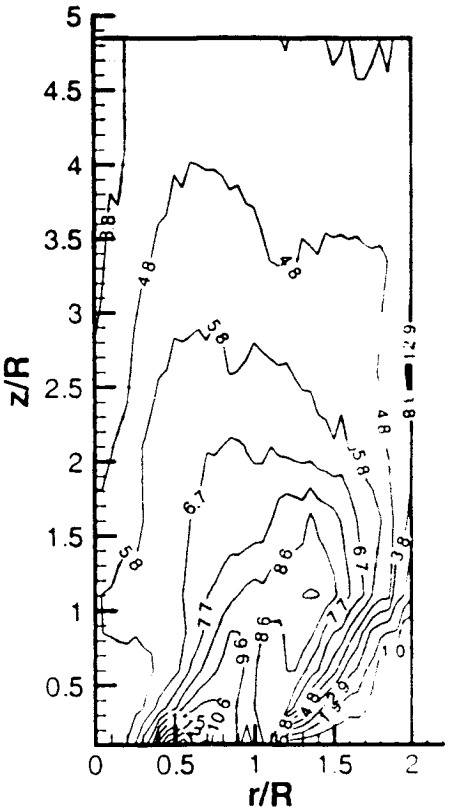


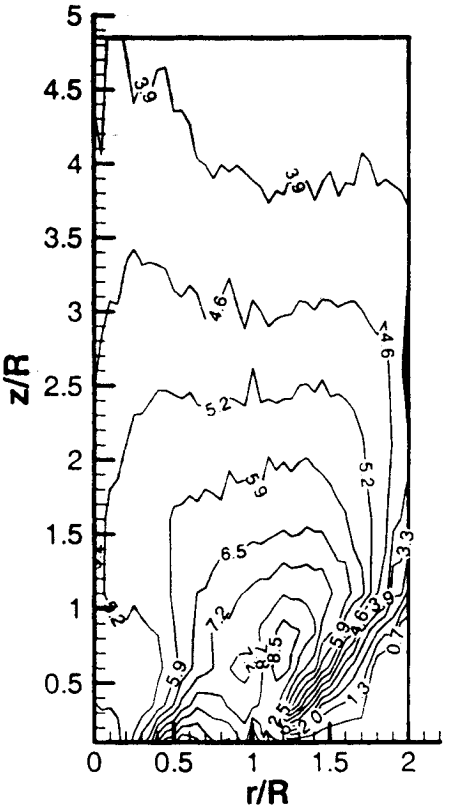

(a) Mean Axial Velocity
Range: -12.96~33.57 m/s




(b) Mean Tangential Velocity
Range: 0~22.3 m/s




(c) RMS Axial Velocity
Range: 0~15.36 m/s




(d) RMS Tangential Velocity
Range: 0~10.43 m/s

Figure 7 Velocity Mapping for half streamwise plane of swirler 504545

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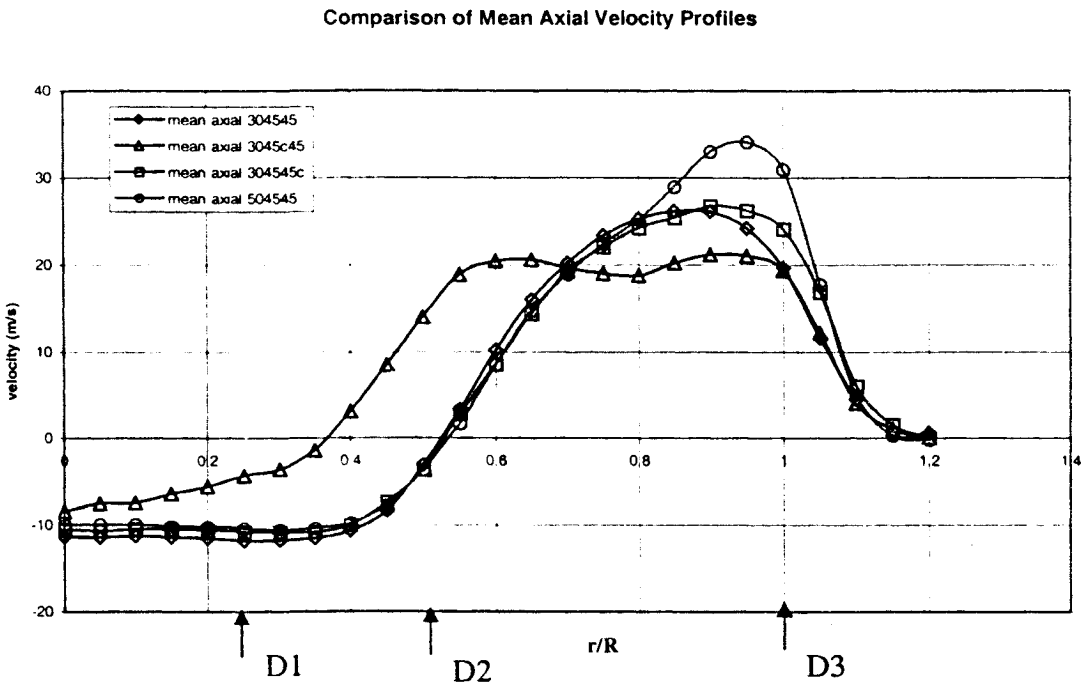
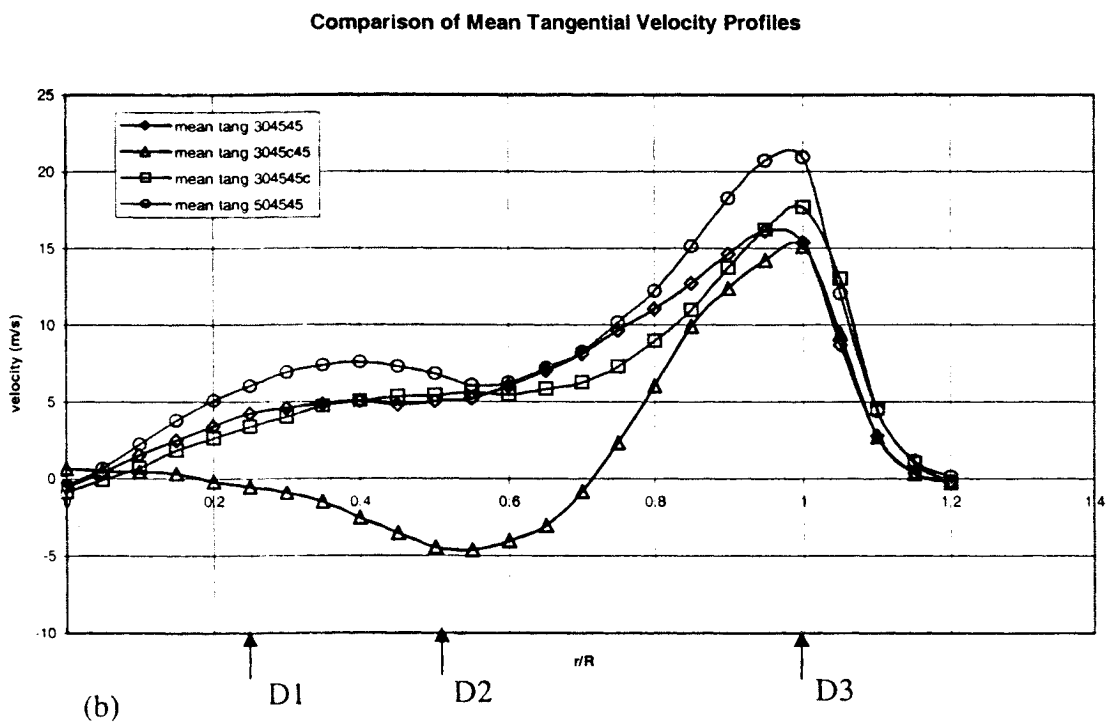



(a)




(b)

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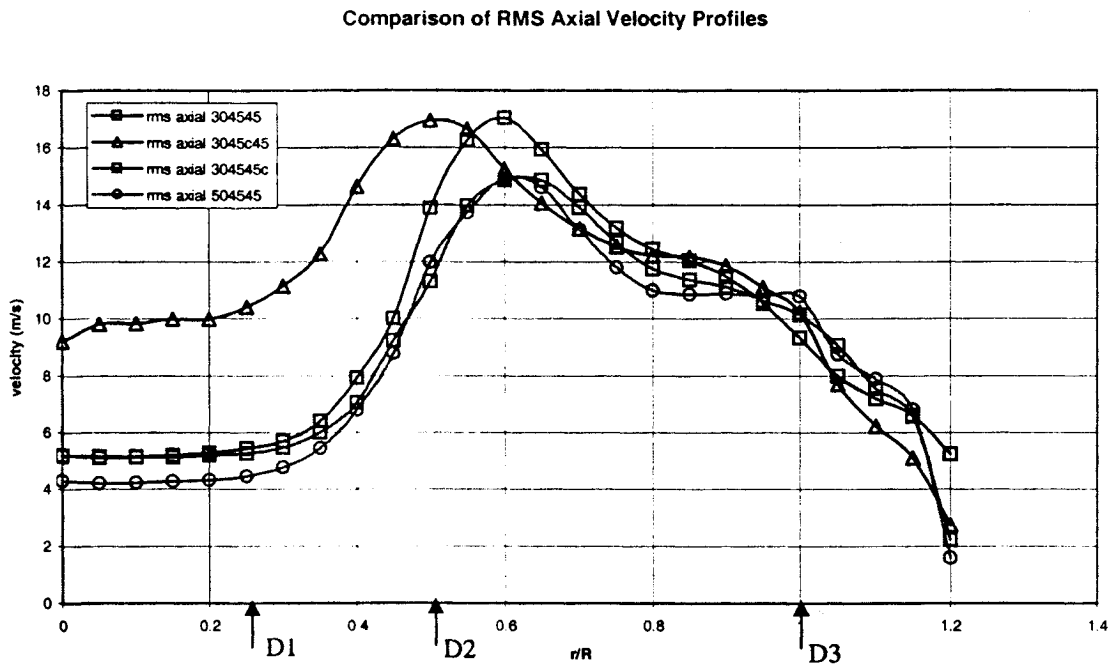
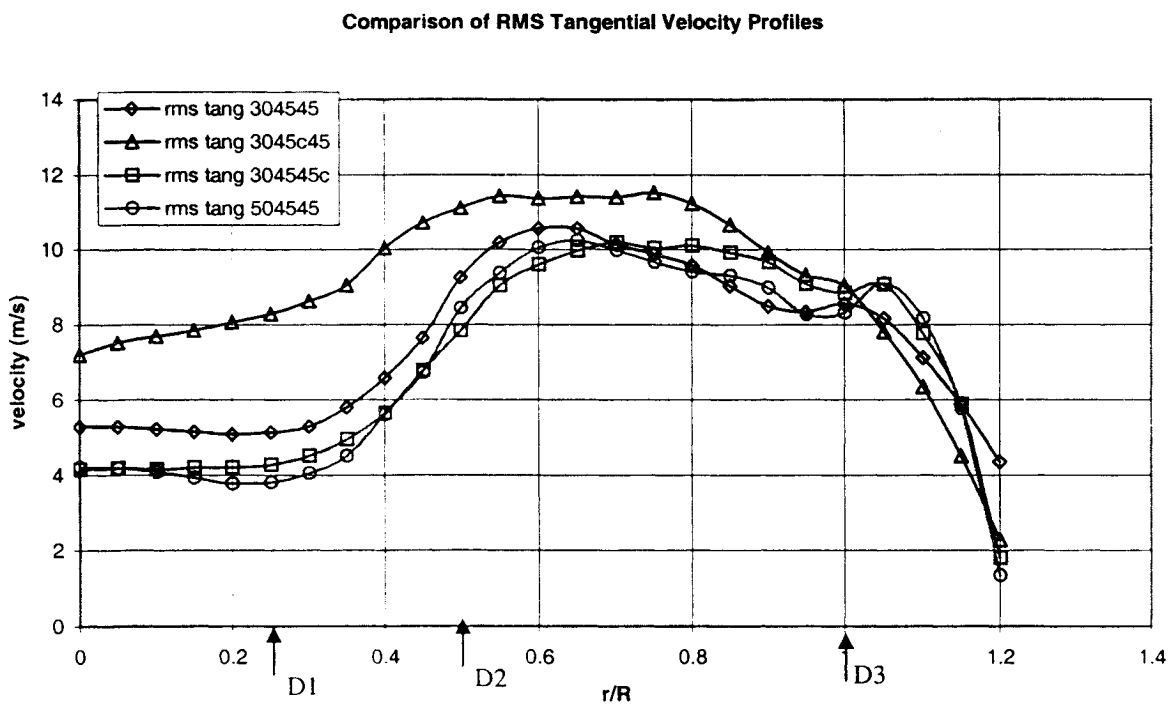



(c)




(d)

Figure 8: Comparison of circumferential averaged velocity profiles for 4 different swirlers at z/R=0.1 (a) Mean Axial (b) Mean Tangential (c) RMS axial (d) RMS tangential

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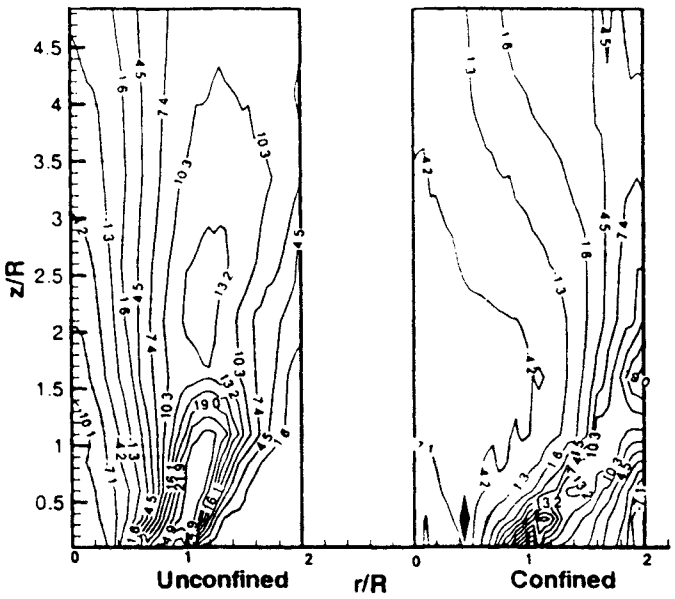
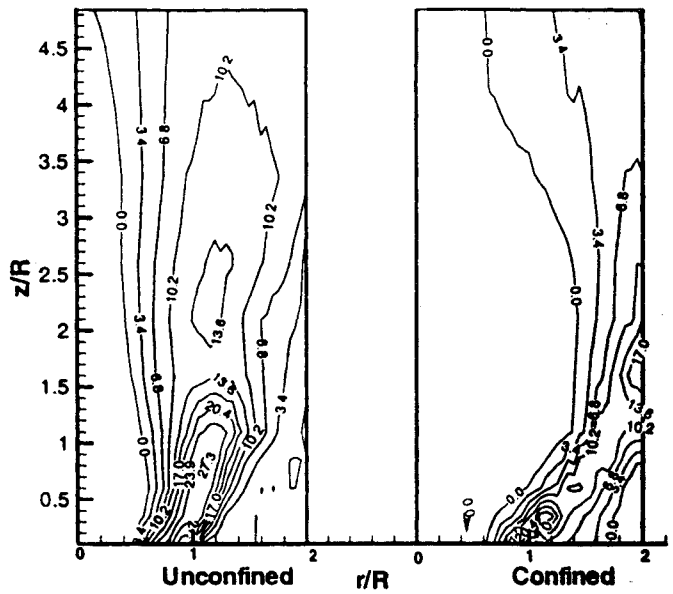
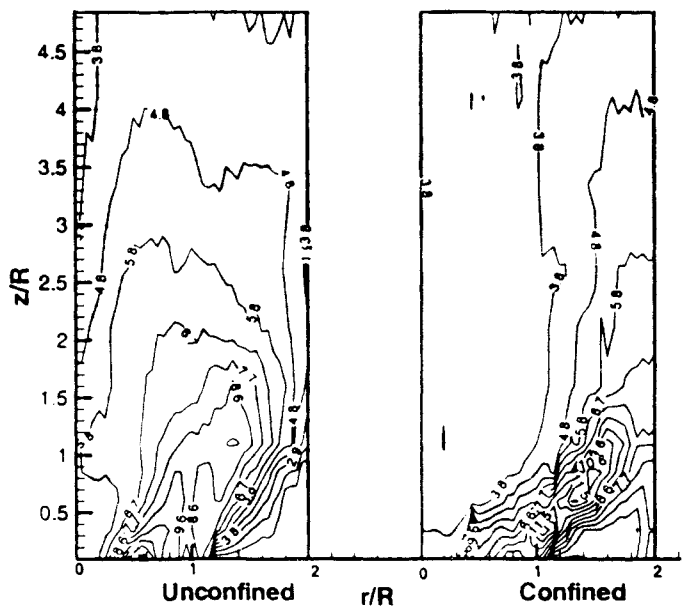



(a) Mean Axial Velocity
Range: -12.9~33.57 m/s




(c) Recirculation Bubble




(b) RMS Axial Velocity
Range: 0~15.36 m/s

Figure 9: Comparison of Axial velocity for swirler 504545 with and without confined tube

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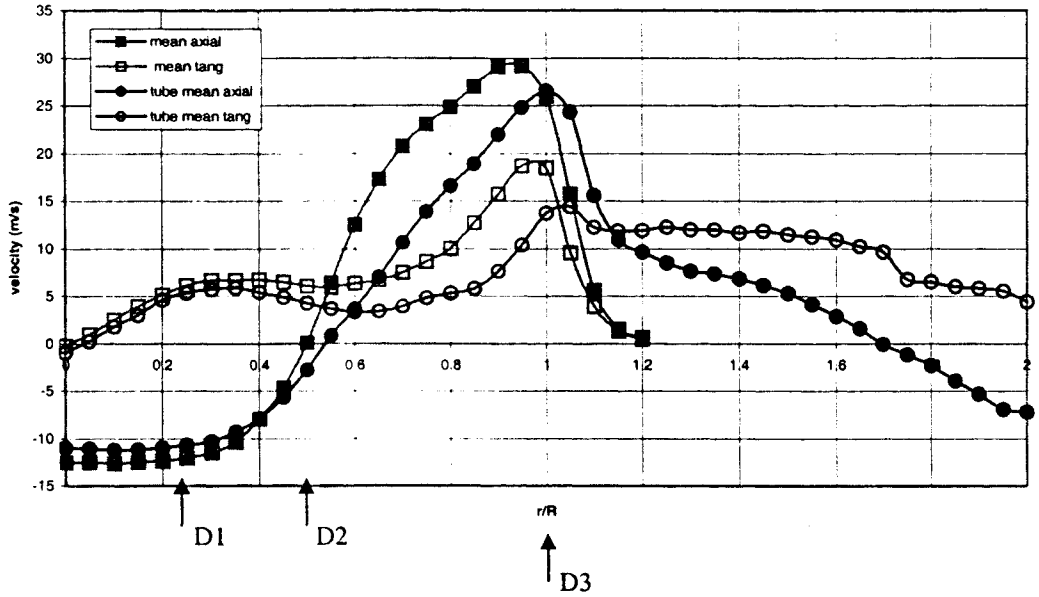
**Comparison of circumferential averaged mean velocity of confined vs unconfined cases**




**Comparison of circumferential averaged RMS velocity confined vs unconfined**

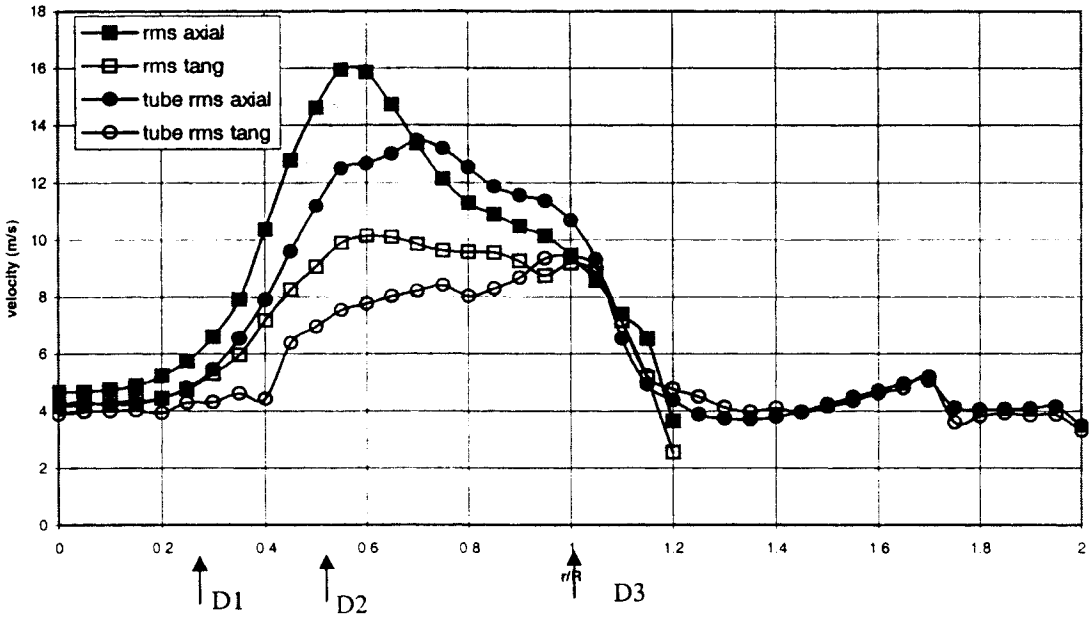



Figure10: Comparison of Circumferential Averaged Velocity profiles for confined vs unconfined case.

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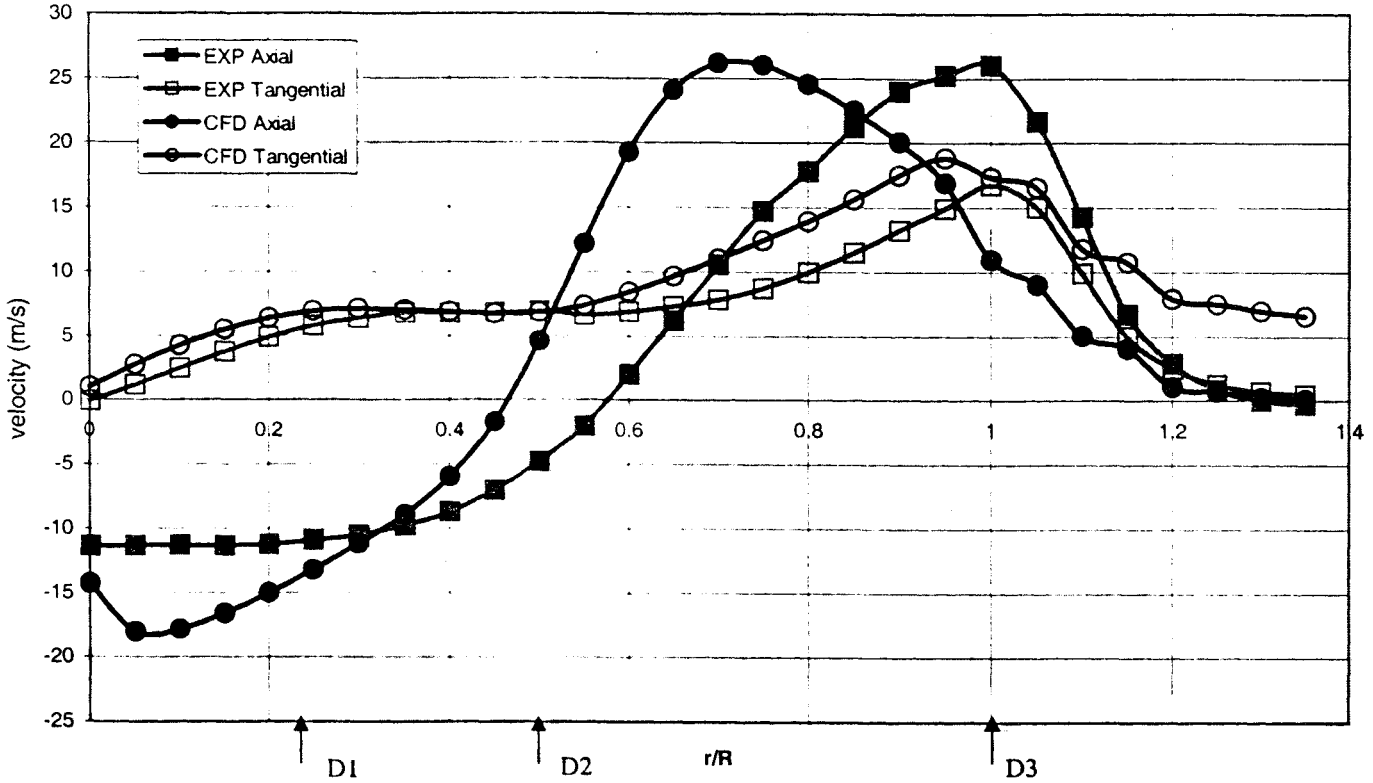
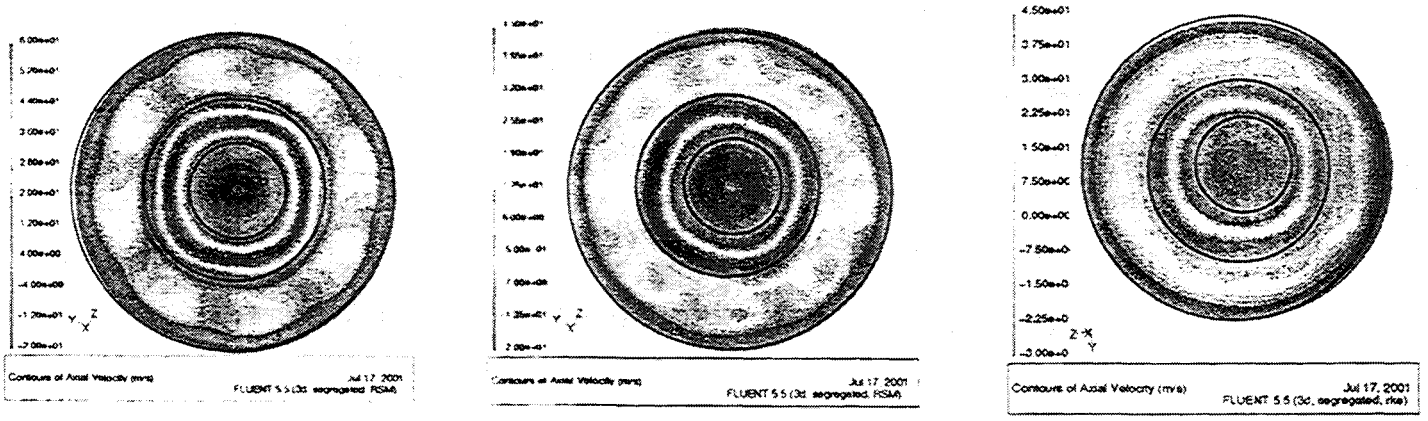



Figure 11: Comparison of Experimental Data with CFD simulation for swirler 304545 at z/R=0.25



Z/R=-0.4 Z/R=-0.2 Z/R=0

Figure 12: Axial Velocity at different z/R locations for swirler 304545